

2016 HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–14

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

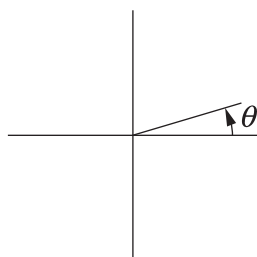
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

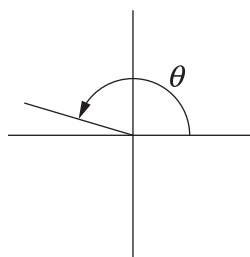
- 1 For the angle θ , $\sin\theta = \frac{7}{25}$ and $\cos\theta = -\frac{24}{25}$.

Which diagram best shows the angle θ ?

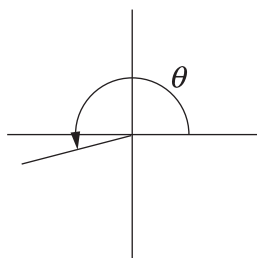
(A)



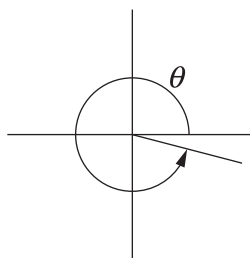
(B)



(C)



(D)

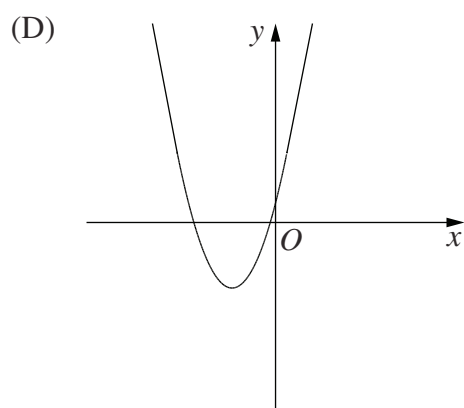
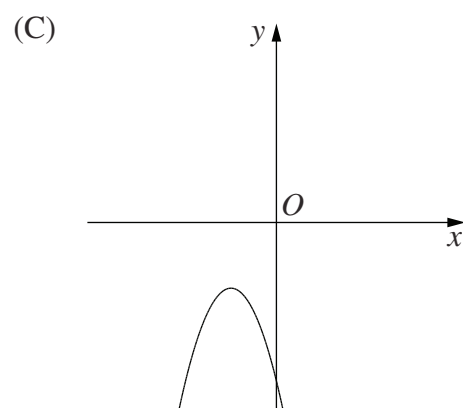
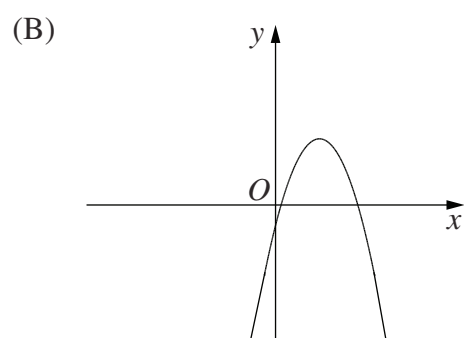
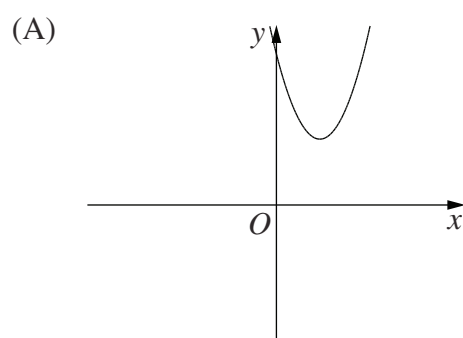


- 2 In a raffle, 30 tickets are sold and there is one prize to be won.

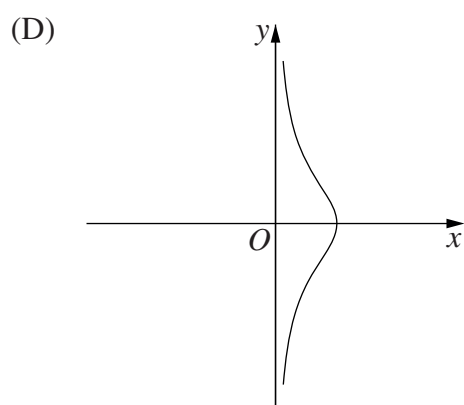
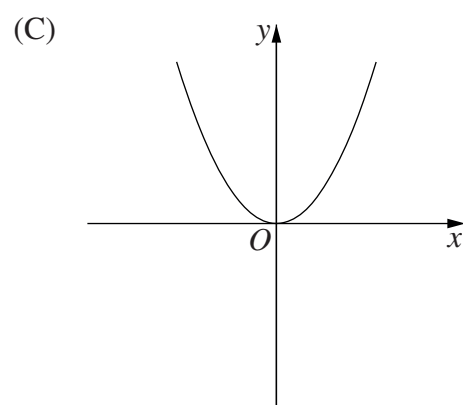
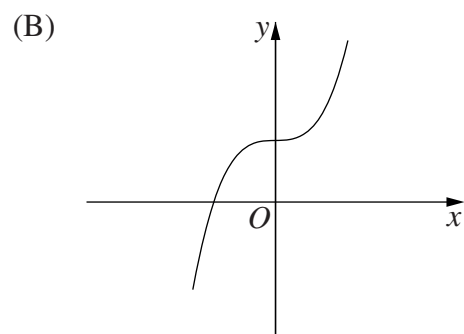
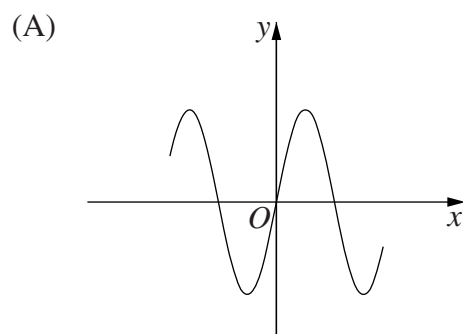
What is the probability that someone buying 6 tickets wins the prize?

- (A) $\frac{1}{30}$
(B) $\frac{1}{6}$
(C) $\frac{1}{5}$
(D) $\frac{1}{4}$

3 Which diagram best shows the graph of the parabola $y = 3 - (x - 2)^2$?



4 Which diagram shows the graph of an odd function?



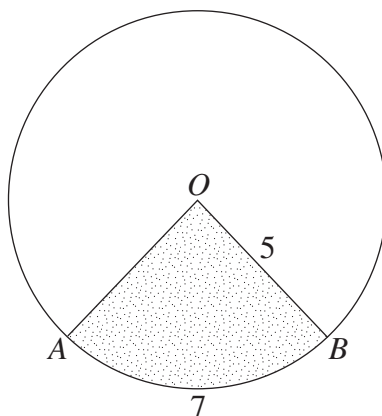
5 What is the derivative of $\ln(\cos x)$?

- (A) $-\sec x$
- (B) $-\tan x$
- (C) $\sec x$
- (D) $\tan x$

6 What is the period of the function $f(x) = \tan(3x)$?

- (A) $\frac{\pi}{3}$
- (B) $\frac{2\pi}{3}$
- (C) 3π
- (D) 6π

7 The circle centred at O has radius 5. Arc AB has length 7 as shown in the diagram.



What is the area of the shaded sector OAB ?

- (A) $\frac{35}{2}$
- (B) $\frac{35}{2}\pi$
- (C) $\frac{125}{14}$
- (D) $\frac{125}{14}\pi$

8 How many solutions does the equation $|\cos(2x)| = 1$ have for $0 \leq x \leq 2\pi$?

- (A) 1
- (B) 3
- (C) 4
- (D) 5

9 What is the value of $\int_{-3}^2 |x+1| dx$?

- (A) $\frac{5}{2}$
- (B) $\frac{11}{2}$
- (C) $\frac{13}{2}$
- (D) $\frac{17}{2}$

10 Which expression is equivalent to $4 + \log_2 x$?

- (A) $\log_2(2x)$
- (B) $\log_2(16+x)$
- (C) $4\log_2(2x)$
- (D) $\log_2(16x)$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

(a) Sketch the graph of $(x - 3)^2 + (y + 2)^2 = 4$. **2**

(b) Differentiate $\frac{x + 2}{3x - 4}$. **2**

(c) Solve $|x - 2| \leq 3$. **2**

(d) Evaluate $\int_0^1 (2x + 1)^3 dx$. **2**

(e) Find the points of intersection of $y = -5 - 4x$ and $y = 3 - 2x - x^2$. **3**

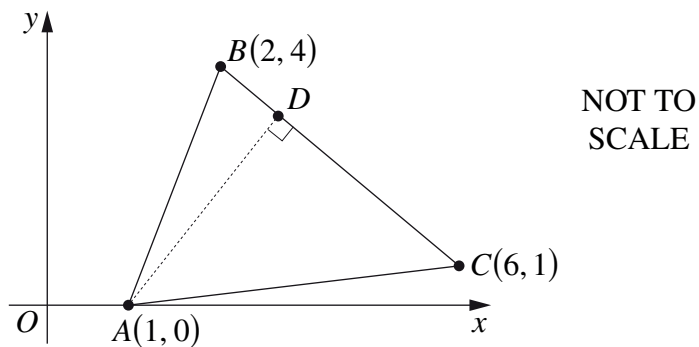
(f) Find the gradient of the tangent to the curve $y = \tan x$ at the point where $x = \frac{\pi}{8}$. **2**

Give your answer correct to 3 significant figures.

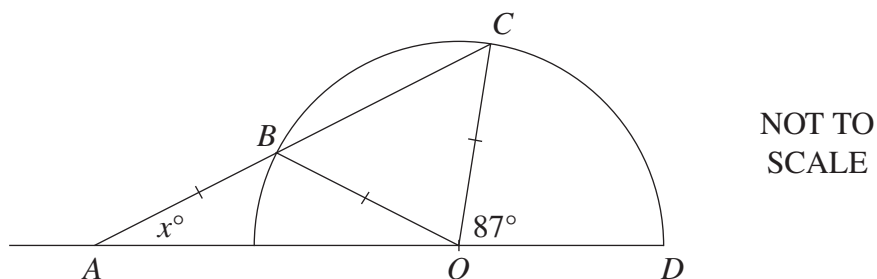
(g) Solve $\sin\left(\frac{x}{2}\right) = \frac{1}{2}$ for $0 \leq x \leq 2\pi$. **2**

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) The diagram shows points $A(1, 0)$, $B(2, 4)$ and $C(6, 1)$. The point D lies on BC such that $AD \perp BC$.



- (i) Show that the equation of BC is $3x + 4y - 22 = 0$. 2
 - (ii) Find the length of AD . 2
 - (iii) Hence, or otherwise, find the area of $\triangle ABC$. 2
- (b) The diagram shows a semicircle with centre O . It is given that $AB = OB$, $\angle COD = 87^\circ$ and $\angle BAO = x^\circ$.



- (i) Show that $\angle CBO = 2x^\circ$, giving reasons. 1
- (ii) Find the value of x , giving reasons. 2

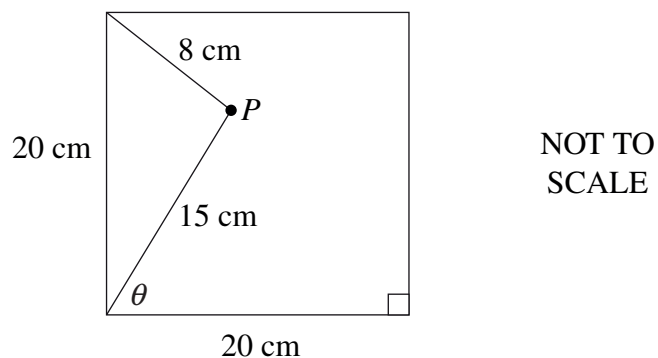
Question 12 continues on page 8

Question 12 (continued)

- (c) Square tiles of side length 20 cm are being used to tile a bathroom. **3**

The tiler needs to drill a hole in one of the tiles at a point P which is 8 cm from one corner and 15 cm from an adjacent corner.

To locate the point P the tiler needs to know the size of the angle θ shown in the diagram.



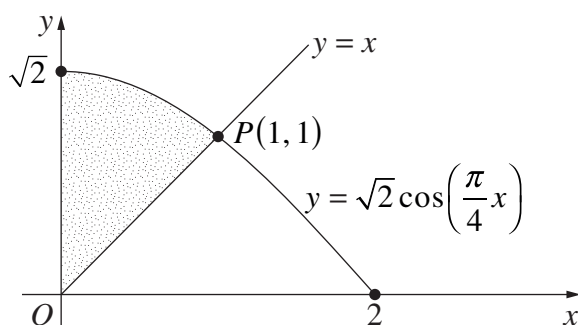
Find the size of the angle θ to the nearest degree.

- (d) (i) Differentiate $y = xe^{3x}$. **1**
- (ii) Hence find the exact value of $\int_0^2 e^{3x}(3+9x)dx$. **2**

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

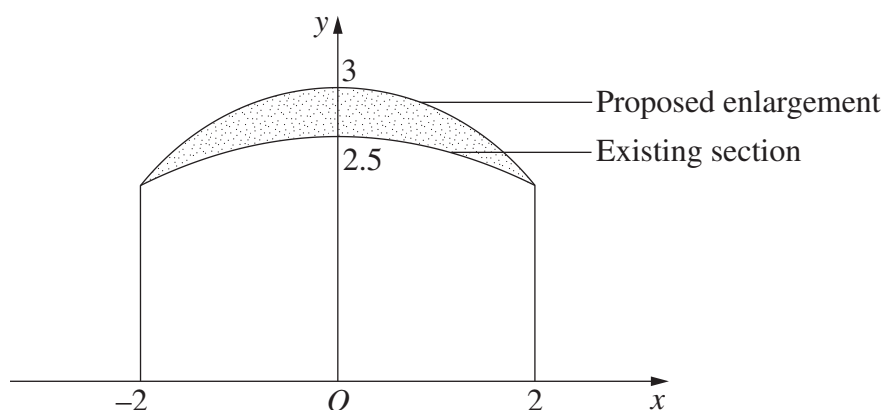
- (a) Consider the function $y = 4x^3 - x^4$.
- (i) Find the two stationary points and determine their nature. 4
 - (ii) Sketch the graph of the function, clearly showing the stationary points and the x and y intercepts. 2
- (b) Consider the parabola $x^2 - 4x = 12y + 8$.
- (i) By completing the square, or otherwise, find the focal length of the parabola. 2
 - (ii) Find the coordinates of the focus. 1
- (c) A radioactive isotope of Curium has a half-life of 163 days. Initially there are 10 mg of Curium in a container.
- The mass $M(t)$ in milligrams of Curium, after t days, is given by
- $$M(t) = Ae^{-kt},$$
- where A and k are constants.
- (i) State the value of A . 1
 - (ii) Given that after 163 days only 5 mg of Curium remain, find the value of k . 2
- (d) The curve $y = \sqrt{2} \cos\left(\frac{\pi}{4}x\right)$ meets the line $y = x$ at $P(1, 1)$, as shown in the diagram. 3



Find the exact value of the shaded area.

Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) The diagram shows the cross-section of a tunnel and a proposed enlargement. **3**



The heights, in metres, of the existing section at 1 metre intervals are shown in Table A.

Table A: Existing heights

x	-2	-1	0	1	2
y	2	2.38	2.5	2.38	2

The heights, in metres, of the proposed enlargement are shown in Table B.

Table B: Proposed heights

x	-2	-1	0	1	2
y	2	2.78	3	2.78	2

Use Simpson's rule with the measurements given to calculate the approximate increase in area.

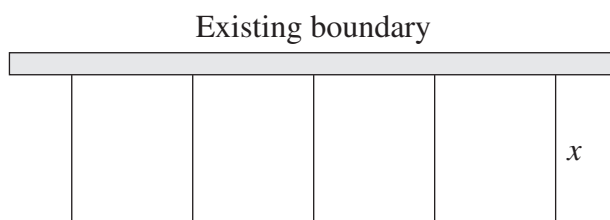
Question 14 continues on page 11

Question 14 (continued)

- (b) A gardener develops an eco-friendly spray that will kill harmful insects on fruit trees without contaminating the fruit. A trial is to be conducted with 100 000 insects. The gardener expects the spray to kill 35% of the insects each day and that exactly 5000 new insects will be produced each day.

The number of insects expected at the end of the n th day of the trial is A_n .

- (i) Show that $A_2 = 0.65(0.65 \times 100\,000 + 5000) + 5000$. 2
- (ii) Show that $A_n = 0.65^n \times 100\,000 + 5000 \frac{(1 - 0.65^n)}{0.35}$. 1
- (iii) Find the expected insect population at the end of the fourteenth day, correct to the nearest 100. 1
- (c) A farmer wishes to make a rectangular enclosure of area 720 m^2 . She uses an existing straight boundary as one side of the enclosure. She uses wire fencing for the remaining three sides and also to divide the enclosure into four equal rectangular areas of width $x \text{ m}$ as shown.

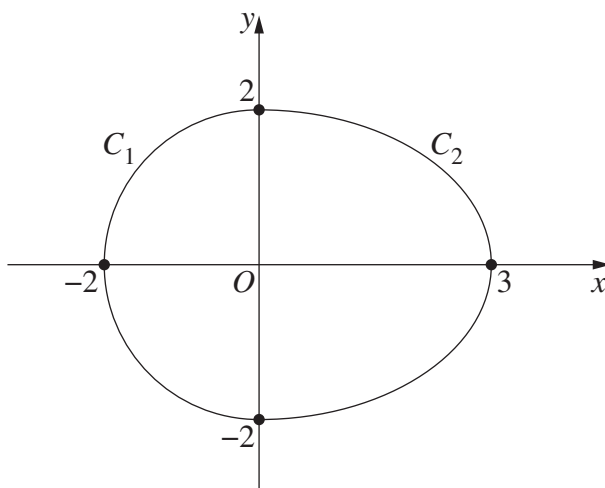


- (i) Show that the total length, $\ell \text{ m}$, of the wire fencing is given by 1
- $$\ell = 5x + \frac{720}{x}.$$
- (ii) Find the minimum length of wire fencing required, showing why this is the minimum length. 3
- (d) By summing the geometric series $1 + x + x^2 + x^3 + x^4$, or otherwise, 2
- find $\lim_{x \rightarrow 1} \frac{x^5 - 1}{x - 1}$.
- (e) Write $\log 2 + \log 4 + \log 8 + \dots + \log 512$ in the form $a \log b$ where a and b are integers greater than 1. 2

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet.

- (a) The diagram shows two curves C_1 and C_2 . The curve C_1 is the semicircle $x^2 + y^2 = 4$, $-2 \leq x \leq 0$. The curve C_2 has equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$, $0 \leq x \leq 3$. **4**



An egg is modelled by rotating the curves about the x -axis to form a solid of revolution.

Find the exact value of the volume of the solid of revolution.

- (b) An eight-sided die is marked with numbers 1, 2, ..., 8. A game is played by rolling the die until an 8 appears on the uppermost face. At this point the game ends. **2**
- (i) Using a tree diagram, or otherwise, explain why the probability of the game ending before the fourth roll is

$$\frac{1}{8} + \frac{7}{8} \times \frac{1}{8} + \left(\frac{7}{8}\right)^2 \times \frac{1}{8}.$$

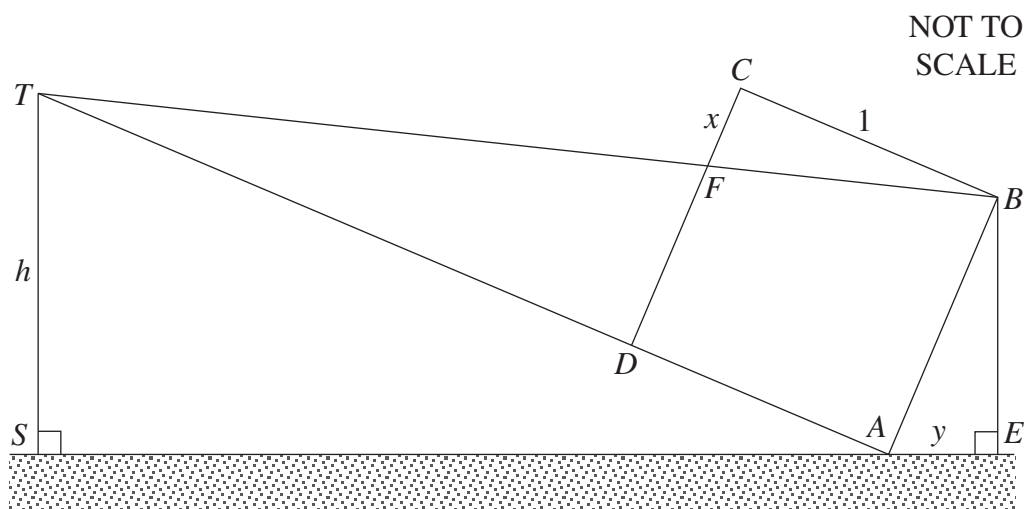
- (ii) What is the smallest value of n for which the probability of the game ending before the n th roll is more than $\frac{3}{4}$? **3**

Question 15 continues on page 13

Question 15 (continued)

- (c) Maryam wishes to estimate the height, h metres, of a tower, ST , using a square, $ABCD$, with side length 1 metre.

She places the point A on the horizontal ground and ensures that the point D lies on the line joining A to the top of the tower T . The point F is the intersection of the line joining B and T and the side CD . The point E is the foot of the perpendicular from B to the ground. Let CF have length x metres and AE have length y metres.



Copy or trace the diagram into your writing booklet.

- | | |
|---|----------|
| (i) Show that $\triangle FCB$ and $\triangle BAT$ are similar. | 2 |
| (ii) Show that $\triangle TSA$ and $\triangle AEB$ are similar. | 2 |
| (iii) Find h in terms of x and y . | 2 |

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) A particle moves in a straight line. Its velocity $v \text{ m s}^{-1}$ at time t seconds is given by

$$v = 2 - \frac{4}{t+1}.$$

- (i) Find the initial velocity. 1
 - (ii) Find the acceleration of the particle when the particle is stationary. 2
 - (iii) By considering the behaviour of v for large t , sketch a graph of v against t for $t \geq 0$, showing any intercepts. 2
 - (iv) Find the exact distance travelled by the particle in the first 7 seconds. 3
- (b) Some yabbies are introduced into a small dam. The size of the population, y , of yabbies can be modelled by the function

$$y = \frac{200}{1 + 19e^{-0.5t}},$$

where t is the time in months after the yabbies are introduced into the dam.

- (i) Show that the rate of growth of the size of the population is 2
$$\frac{1900e^{-0.5t}}{(1 + 19e^{-0.5t})^2}.$$
- (ii) Find the range of the function y , justifying your answer. 2
- (iii) Show that the rate of growth of the size of the population can be rewritten as 1

$$\frac{y}{400}(200 - y).$$

- (iv) Hence, find the size of the population when it is growing at its fastest rate. 2

End of paper

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REFERENCE SHEET

– Mathematics –

– Mathematics Extension 1 –

– Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

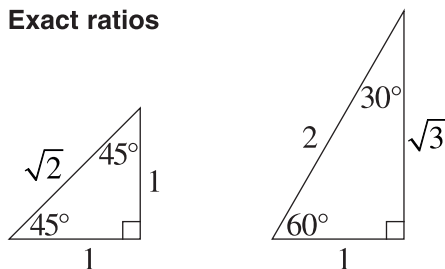
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

tangent: $y = tx - at^2$

normal: $x + ty = at^3 + 2at$

At (x_1, y_1) ,

tangent: $xx_1 = 2a(y + y_1)$

normal: $y - y_1 = -\frac{2a}{x_1}(x - x_1)$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2016 HSC Mathematics Marking Guidelines

Section I

Multiple-choice Answer Key

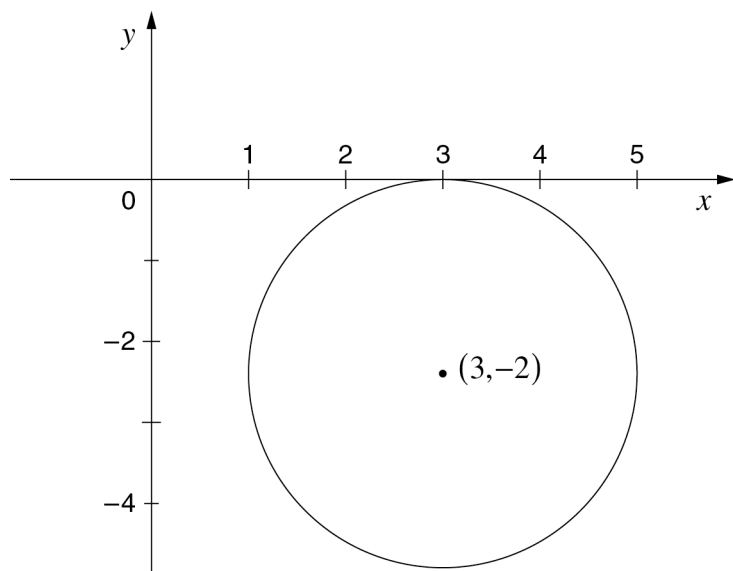
Question	Answer
1	B
2	C
3	B
4	A
5	B
6	A
7	A
8	D
9	C
10	D

Section II

Question 11 (a)

Criteria	Marks
• Provides correct sketch	2
• Identifies radius, or equivalent merit	1

Sample answer:



Question 11 (b)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use quotient rule, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{d}{dx} \left(\frac{x+2}{3x-4} \right) &= \frac{(3x-4)(1) - (x+2)(3)}{(3x-4)^2} \\
 &= \frac{3x-4-3x-6}{(3x-4)^2} \\
 &= \frac{-10}{(3x-4)^2}
 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Establishes that $x \leq 5$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 |x - 2| &\leq 3 \\
 \therefore -3 &\leq x - 2 \leq 3 \\
 -1 &\leq x \leq 5
 \end{aligned}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Correct primitive, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \int_0^1 (2x + 1)^3 dx &= \left[\frac{1}{2} \times \frac{1}{4} (2x + 1)^4 \right]_0^1 \\
 &= \frac{1}{8} (3)^4 - \frac{1}{8} (1)^4 \\
 &= \frac{81}{8} - \frac{1}{8} \\
 &= \frac{80}{8} \\
 &= 10
 \end{aligned}$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Obtains $x^2 - 2x - 8 = 0$ and solves for x , or equivalent merit	2
• Attempts to eliminate x or y , or equivalent merit	1

Sample answer:

$$\left. \begin{array}{l} y = -5 - 4x \\ y = 3 - 2x - x^2 \end{array} \right\} \begin{array}{l} \textcircled{1} \\ \textcircled{2} \end{array}$$

Subs $\textcircled{1}$ into $\textcircled{2}$

$$-5 - 4x = 3 - 2x - x^2$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$\therefore x = 4 \text{ or } -2$$

$$\text{Subst } x = 4 \text{ into } \textcircled{1} \Rightarrow y = -5 - 4(4) = -21$$

$$\text{Subst } x = -2 \text{ into } \textcircled{1} \Rightarrow y = -5 - 4(-2) = 3$$

 $\therefore$ points of intersection are $(4, -21)$ and $(-2, 3)$.**Question 11 (f)**

Criteria	Marks
• Provides correct solution	2
• Obtains $\sec^2 \frac{\pi}{8}$, or equivalent merit	1

Sample answer:

$$y = \tan x$$

$$\frac{dy}{dx} = \sec^2 x$$

$$\text{When } x = \frac{\pi}{8}, \frac{dy}{dx} = \sec^2 \left(\frac{\pi}{8} \right)$$

$$= \frac{1}{\cos^2 \frac{\pi}{8}}$$

$$\approx 1.17$$

 $\therefore$ Gradient of tangent is 1.17.

Question 11 (g)

Criteria	Marks
• Provides correct solution	2
• Obtains one correct answer, or equivalent merit	1

Sample answer:

$$\sin\left(\frac{x}{2}\right) = \frac{1}{2} \quad \text{for} \quad 0 \leq x \leq 2\pi$$

$$\text{Note} \quad 0 \leq \frac{x}{2} \leq \pi$$

$$\therefore \left(\frac{x}{2}\right) = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$

Question 12 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains correct slope, or equivalent merit	1

Sample answer:

$$\frac{y-4}{x-2} = \frac{1-4}{6-2}$$

$$\frac{y-4}{x-2} = \frac{-3}{4}$$

$$4(y-4) = -3(x-2)$$

$$4y - 16 = -3x + 6$$

$$3x + 4y - 22 = 0$$

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to calculate perpendicular distance of a point from a line, or equivalent merit	1

Sample answer:

$$BC : 3x + 4y - 22 = 0$$

$$\begin{aligned}\therefore AD &= \frac{|3(1) + 4(0) - 22|}{\sqrt{3^2 + 4^2}} \\ &= \frac{|-19|}{\sqrt{25}} \\ &= \frac{19}{5}\end{aligned}$$

Question 12 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Finds distance from B to C , or equivalent merit	1

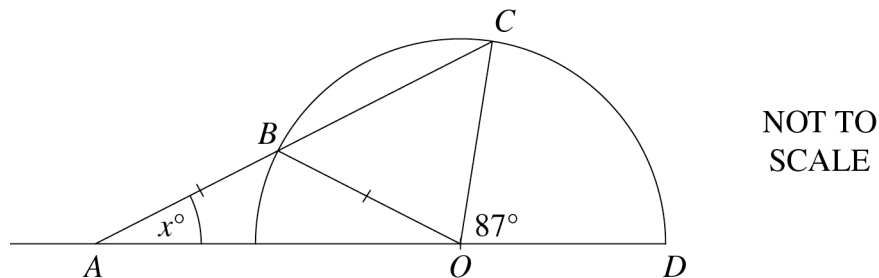
Sample answer:

$$\begin{aligned}BC &= \sqrt{(2-6)^2 + (4-1)^2} \\ &= \sqrt{16+9} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

$$\begin{aligned}\therefore \text{Area } \triangle ABC &= \frac{1}{2} \times 5 \times \frac{19}{5} \\ &= \frac{19}{2} \text{ square units}\end{aligned}$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

$$\angle BOA = x^\circ \quad (\angle\text{s opposite equal sides in } \triangle ABO)$$

$$\therefore \angle CBO = 2x^\circ \quad (\text{exterior } \angle \text{ of } \triangle ABO \text{ equal to sum of two opposite interior } \angle\text{s})$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the fact that triangle OBC is isosceles, or equivalent merit	1

Sample answer:

$$OC = OB \text{ (radii)}$$

$$\angle BCO = \angle CBO \text{ (}\angle\text{s opposite equal sides in } \triangle CBO\text{)}$$

$$= 2x^\circ$$

Then $\angle COD = \angle CAO + \angle BCO$ (exterior $\angle$ of $\triangle CAO$ equals sum of two opposite interior $\angle$ s)

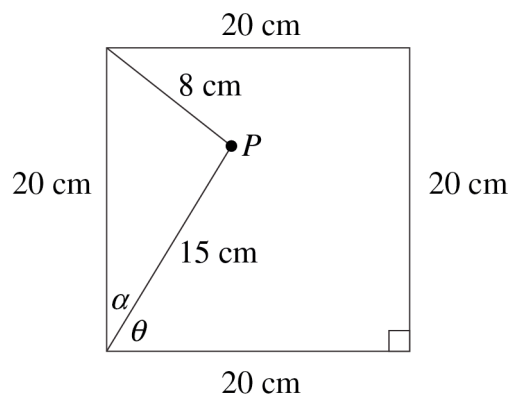
$$\therefore 87 = x + 2x$$

$$87 = 3x$$

$$x = 29$$

Question 12 (c)

Criteria	Marks
• Provides correct solution	3
• Substitutes correctly into cosine rule, and finds angle or equivalent merit	2
• Identifies the complementary angle to θ , or equivalent merit	1

Sample answer:

$$\begin{aligned}\cos \alpha &= \frac{15^2 + 20^2 - 8^2}{2 \times 20 \times 15} \\ &= \frac{561}{600} \\ \alpha &\approx 20.77^\circ \\ \theta &\approx 69^\circ \text{ to nearest degree.}\end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct derivative	1

Sample answer:

$$y = xe^{3x}$$

$$\begin{aligned}\frac{dy}{dx} &= 3xe^{3x} + e^{3x} \\ &= e^{3x}(1 + 3x)\end{aligned}$$

Question 12 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use part (i) or equivalent merit	1

Sample answer:

Find exact value of $\int_0^2 e^{3x}(3+9x)dx$

$$\begin{aligned}
 \int_0^2 e^{3x}(3+9x)dx &= \int_0^2 3e^{3x}(1+3x)dx \\
 &= 3 \int_0^2 e^{3x}(1+3x)dx \\
 &= 3 \left[xe^{3x} \right]_0^2 \\
 &= 3 \left[(2e^6) - (0e^0) \right] \\
 &= 6e^6
 \end{aligned}$$

Question 13 (a) (i)

Criteria	Marks
• Provides correct solution	4
• Finds the x -values at which the stationary points occur and verifies the maximum turning point, or equivalent merit	3
• Finds the x -values at which the stationary points occur, or equivalent merit	2
• Attempts to solve $\frac{dy}{dx} = 0$, or equivalent merit	1

Sample answer:

$$y = 4x^3 - x^4$$

Find stationary points and determine nature.

$$\frac{dy}{dx} = 12x^2 - 4x^3$$

$$\begin{aligned} \text{Need } \frac{dy}{dx} &= 0 & 12x^2 - 4x^3 &= 0 \\ & & 4x^2(3 - x) &= 0 \\ & & \text{when } x = 0 \text{ or } x = 3. \end{aligned}$$

$$\begin{aligned} \text{when } x = 0 & \quad y = 0 \\ \text{when } x = 3 & \quad y = 27 \end{aligned}$$

Checking the gradients for

$$x = 0, \quad \begin{array}{c|c|c|c} x & 0^- & 0 & 0^+ \\ \hline y' & +ve & 0 & +ve \end{array}$$

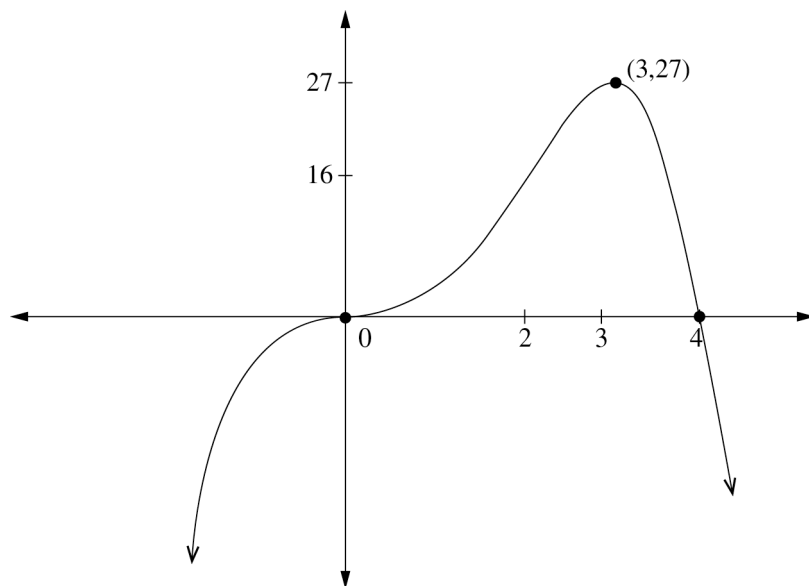
 $\therefore$ horizontal point of inflexion at $x = 0$, ie at $(0, 0)$

$$x = 3, \quad \begin{array}{c|c|c|c} x & 3^+ & 3 & 3^- \\ \hline y' & +ve & 0 & -ve \end{array}$$

 $\therefore$ local maximum at $x = 3$, ie at $(3, 27)$

Question 13 (a) (ii)

Criteria	Marks
• Correct solution	2
• Locates the stationary points on the sketch of the curve, or equivalent merit	1

Sample answer:**Question 13 (b) (i)**

Criteria	Marks
• Provides correct solution	2
• Attempts to complete the square, or equivalent merit	1

Sample answer:

$$x^2 - 4x = 12y + 8$$

$$x^2 - 4x + 4 = 12y + 12$$

$$x^2 - 4x + 4 = 12(y + 1)$$

$$(x - 2)^2 = 4(3)(y + 1)$$

$$\therefore \text{focal length} = 3$$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer: $\therefore$ focus (2,2)**Question 13 (c) (i)**

Criteria	Marks
• Provides correct answer	1

Sample answer: $A = 10$ **Question 13 (c) (ii)**

Criteria	Marks
• Provides correct solution	2
• Obtains $5 = 10e^{-163k}$, or equivalent merit	1

Sample answer:

$$M(t) = 10e^{-kt}$$

$$M(163) = 5$$

$$\therefore 5 = 10e^{-163k}$$

$$\frac{1}{2} = e^{-163k}$$

$$-163k = \log_e \left(\frac{1}{2} \right)$$

$$k = \frac{\log_e \left(\frac{1}{2} \right)}{-163}$$

$$\approx 0.004252436$$

Question 13 (d)

Criteria	Marks
• Provides correct solution	3
• Finds the area under the cosine curve, or equivalent merit	2
• Finds the area under $y = x$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \text{Area} &= \int_0^1 \sqrt{2} \cos\left(\frac{\pi}{4}x\right) dx - \frac{1}{2} \times 1 \times 1 \\
 &= \sqrt{2} \times \frac{4}{\pi} \left[\sin\left(\frac{\pi}{4}x\right) \right]_0^1 - \frac{1}{2} \\
 &= \frac{4\sqrt{2}}{\pi} \left(\sin \frac{\pi}{4} - \sin 0 \right) - \frac{1}{2} \\
 &= \frac{4\sqrt{2}}{\pi} \left(\frac{1}{\sqrt{2}} - 0 \right) - \frac{1}{2} \\
 &= \frac{4}{\pi} - \frac{1}{2} \text{ square units} \\
 &\text{or } \frac{8 - \pi}{2\pi}
 \end{aligned}$$

Question 14 (a)

Criteria	Marks
• Provides correct solution	3
• Attempts to find a difference in areas or equivalent merit	2
• Applies Simpson's Rule to the existing heights or equivalent merit	1

Sample answer:

The increase in area can be approximated using Simpson's Rule.

$$\begin{aligned}
 \left. \begin{array}{l} \text{Area increase} \\ \text{(Simpson, 5 values)} \end{array} \right\} &= \frac{1}{3} [2 \times (2 - 2) + 2 \times (3 - 2.5) + 8 \times (2.78 - 2.38)] \\
 &= \frac{1}{3} [0 + 1 + 3.2] \\
 &\approx 1.4 \text{ m}^2
 \end{aligned}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains a correct expression for A_1 , or equivalent merit	1

Sample answer:

$$A_0 = 100\,000$$

$$A_1 = 0.65 \times 100\,000 + 5000$$

$$A_2 = 0.65 A_1 + 5000$$

$$= 0.65(0.65 \times 100\,000 + 5000) + 5000$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$A_2 = 0.65^2 \times 100\,000 + 0.65 \times 5000 + 5000$$

$$A_n = 0.65^n \times 100\,000 + 5000(0.65^{n-1} + 0.65^{n-2} + \dots + 1)$$

$$S_n = \frac{1(1 - 0.65^n)}{0.35}$$

$$A_n = 0.65^n \times 100\,000 + 5000 \frac{(1 - 0.65^n)}{0.35}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$A_{14} = 0.65^{14} \times 100\,000 + 5000 \frac{(1 - 0.65^{14})}{0.35}$$

$$= 14\,491.7\Box$$

$$\approx 14\,500$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

If w is the total width then

$$\text{Area} = 720 = x \times w$$

$$w = \frac{720}{x}$$

$$\text{Perimeter} = 5 \times x + w$$

$$= 5x + \frac{720}{x}$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds length at stationary point, or equivalent merit	2
• Finds $\frac{d\ell}{dx}$, or equivalent merit	1

Sample answer:

Stationary points occur when

$$\begin{aligned} 0 &= \frac{d\ell}{dx} \\ &= 5 - \frac{720}{x^2} \end{aligned}$$

$$\frac{720}{x^2} = 5$$

$$x^2 = 144$$

$$x = 12 \quad (x \text{ is length so ignore } -12)$$

$$\frac{d^2\ell}{dx^2} = \frac{1440}{x^3}$$

$$\text{at } x = 12 \quad \frac{d^2\ell}{dx^2} = \frac{1440}{12^3} > 0$$

so minimum at $x = 12$

$$\begin{aligned} \ell &= 5 \times 12 + \frac{720}{12} \\ &= 120 \end{aligned}$$

Question 14 (d)

Criteria	Marks
• Provides correct solution	2
• Sums the series, or equivalent merit	1

Sample answer:

$$1 + x + x^2 + x^3 + x^4 = \frac{1 - x^5}{1 - x} \quad a = 1, r = x$$

$$= \frac{x^5 - 1}{x - 1} \quad \text{for } x \neq 1$$

$$\lim_{x \rightarrow 1} \frac{x^5 - 1}{x - 1} = \lim_{x \rightarrow 1} 1 + x + x^2 + x^3 + x^4$$

$$= 5$$

Question 14 (e)

Criteria	Marks
• Provides correct solution	2
• Expresses terms involving powers of 2, or equivalent merit	1

Sample answer:

$$\log 2 + \log 4 + \log 8 + \cdots + \log 512$$

$$= \log 2 + \log 2^2 + \log 2^3 + \cdots + \log 2^9$$

$$= \log 2 + 2\log 2 + 3\log 2 + \cdots + 9\log 2$$

$$= \frac{9}{2}(\log 2 + 9\log 2)$$

$$= \frac{9}{2} \times 10\log 2$$

$$= 45\log 2$$

Question 15 (a)

Criteria	Marks
• Provides correct solution	4
• Evaluates the correct integral for the volume when C_2 is revolved, or equivalent merit	3
• Recognises the sum of two volumes is required and makes progress with both, or equivalent merit	2
• Attempts to find a volume using integration, or equivalent merit	1

Sample answer:

Volume of C_1 rotated:

$$V_1 = \frac{1}{2} \times \frac{4}{3} \pi 2^3$$

$$= \frac{16\pi}{3}$$

Volume of C_2 rotated:

$$V_2 = \pi \int_0^3 y^2 dx$$

$$= \pi \int_0^3 4 \left(1 - \frac{x^2}{9} \right) dx$$

$$= 4\pi \left[x - \frac{x^3}{27} \right]_0^3$$

$$= 4\pi \left(3 - \frac{27}{27} \right)$$

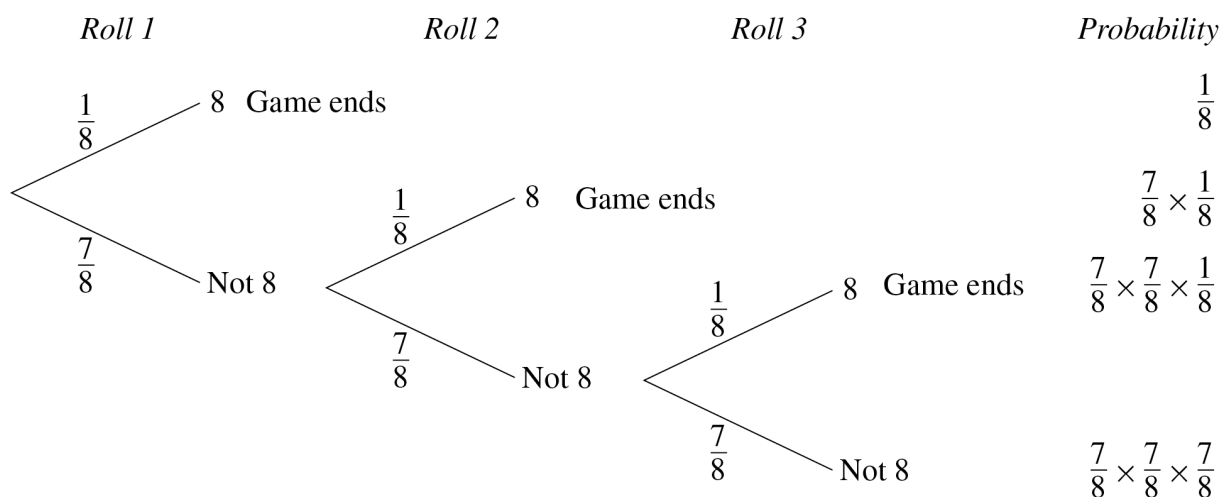
$$= 8\pi$$

$$\therefore V = \frac{16\pi}{3} + 8\pi$$

$$= \frac{40\pi}{3}$$

Question 15 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Provides correct tree diagram, or equivalent merit	1

Sample answer:

$$\therefore P(\text{game ends before 4th roll}) = \frac{1}{8} + \frac{7}{8} \times \frac{1}{8} + \left(\frac{7}{8}\right)^2 \times \frac{1}{8}$$

Alternative solution:

$$P(\text{ends after 1 roll}) = \frac{1}{8}$$

$$P(\text{ends after 2 rolls}) = \frac{7}{8} \times \frac{1}{8}$$

$$P(\text{ends after 3 rolls}) = \left(\frac{7}{8}\right)^2 \times \frac{1}{8}$$

$$\therefore P(\text{ends before 4th roll}) = \frac{1}{8} + \frac{7}{8} \times \frac{1}{8} + \left(\frac{7}{8}\right)^2 \times \frac{1}{8}$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds an equation or inequality of probabilities that can be solved for n , or equivalent merit	2
• Finds, as a series, the probability that the game ends before the n th roll, or equivalent merit	1

Sample answer:

P (ends before n th roll)

$$\begin{aligned}
 &= \frac{1}{8} + \frac{7}{8} \times \frac{1}{8} + \left(\frac{7}{8}\right)^2 \times \frac{1}{8} + \left(\frac{7}{8}\right)^3 \times \frac{1}{8} + \cdots + \left(\frac{7}{8}\right)^{n-2} \times \frac{1}{8} \\
 &= \frac{1}{8} \left[1 + \frac{7}{8} + \left(\frac{7}{8}\right)^2 + \left(\frac{7}{8}\right)^3 + \cdots + \left(\frac{7}{8}\right)^{n-2} \right] \\
 &= \frac{1}{8} \times \frac{\left(\left(\frac{7}{8}\right)^{n-1} - 1\right)}{\left(\frac{7}{8}\right) - 1} \\
 &= 1 - \left(\frac{7}{8}\right)^{n-1}
 \end{aligned}$$

$$\text{Let } 1 - \left(\frac{7}{8}\right)^{n-1} = \frac{3}{4}$$

$$\left(\frac{7}{8}\right)^{n-1} = \frac{1}{4}$$

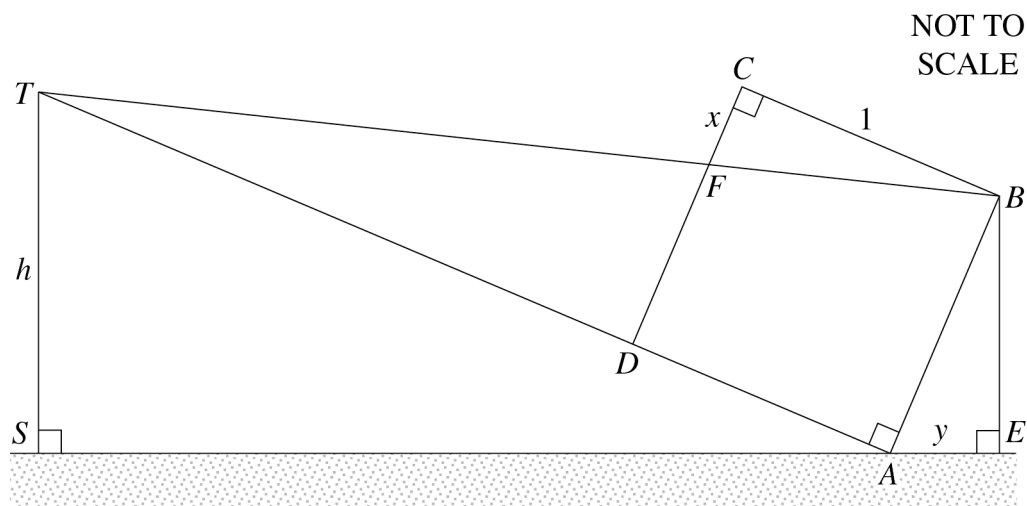
$$(n-1)\log\left(\frac{7}{8}\right) = \log\left(\frac{1}{4}\right)$$

$$\begin{aligned}
 n &= 1 + \frac{\log \frac{1}{4}}{\log \frac{7}{8}} \\
 &\approx 11.3817...
 \end{aligned}$$

$\therefore$ For probability of more than $\frac{3}{4}$, we require $n = 12$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Identifies one pair of equal angles, giving reason(s)	1

Sample answer:In $\triangle s FCB, BAT$ $\angle FCB = \angle BAT$ (both 90° angles in square $ABCD$)Now $AB \parallel DC$ (opposite sides of a square) $\therefore \angle CFB = \angle ABT$ (alternate $\angle s$, $AB \parallel DC$) $\therefore \triangle FCB \parallel \triangle BAT$ (2 pairs of equal $\angle s$).**Question 15 (c) (ii)**

Criteria	Marks
• Provides correct solution	2
• Shows that $\angle TAS$ and $\angle BAE$ are complementary, or equivalent merit	1

Sample answer: $\angle TSA = \angle BAD = \angle AEB = 90^\circ$ $\angle TAS + \angle BAE = 90^\circ$ ($\angle$ sum, straight line SAE) $\angle ABE + \angle BAE = 90^\circ$ ($\angle$ sum $\triangle ABE$) $\therefore \angle TAS = \angle ABE$ In $\triangle s TSA, AEB$ $\angle TAS = \angle ABE$ (above) $\angle TSA = \angle AEB = 90^\circ$ (given) $\therefore \triangle TSA \parallel \triangle AEB$ (2 pairs of equal $\angle s$).

Question 15 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains $AT = \frac{1}{x}$, or equivalent merit	1

Sample answer:

$$\frac{TS}{AE} = \frac{TA}{AB} \quad (\text{matching sides in similar } \triangle\text{s})$$

$$\frac{h}{y} = \frac{TA}{1}$$

$$\therefore h = y.TA$$

Also $\frac{TA}{BC} = \frac{BA}{FC} \quad (\text{matching sides in similar } \triangle\text{s})$

$$\frac{TA}{1} = \frac{1}{x}$$

$$\therefore TA = \frac{1}{x}$$

Hence $h = y \times \frac{1}{x}$

$$\therefore h = \frac{y}{x}$$

Question 16 (a) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:When $t = 0$

$$v = 2 - \frac{4}{0+1} = -2$$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds the value of t for which $v = 0$, or equivalent merit	1

Sample answer:

The particle is stationary when $v = 0$.

$$\text{So } v = 0 \Rightarrow 0 = 2 - \frac{4}{t+1}$$

$$\frac{4}{t+1} = 2$$

$$4 = 2(t+1)$$

$$4 = 2t + 2$$

$$2 = 2t$$

$$1 = t$$

So particle is stationary when $t = 1$.

$$\text{acceleration} = \frac{dv}{dt}$$

$$\frac{dv}{dt} = -4(t+1)^{-2} \times -1$$

$$= \frac{4}{(t+1)^2}$$

$$\text{when } t = 1 \quad \frac{dv}{dt} = \frac{4}{(1+1)^2} = \frac{4}{4} = 1$$

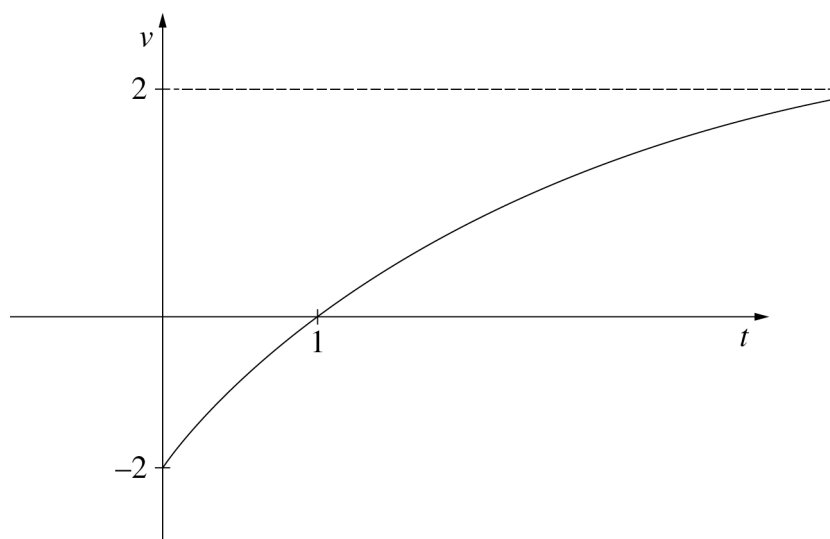
acceleration is 1 ms^{-2} when particle is stationary.

Question 16 (a) (iii)

Criteria	Marks
• Provides correct graph	2
• Describes the behaviour of v for large t , or equivalent merit	1

Sample answer:

As $t \rightarrow \infty, \frac{4}{t+1} \rightarrow 0$
 $\Rightarrow v \rightarrow 2$



Question 16 (a) (iv)

Criteria	Marks
• Provides correct solution	3
• Correctly evaluates $\int_1^7 v dt$, or equivalent merit	2
• Recognises the particle changes direction at $t = 1$, or equivalent merit	1

Sample answer:

Particle changes direction when $t = 1$ travelling in negative direction for $0 \leq t < 1$, so

$$\begin{aligned}
 \text{Distance travelled} &= -\int_0^1 v dt + \int_1^7 v dt \\
 &= -\int_0^1 2 - \frac{4}{t+1} dt + \int_1^7 2 - \frac{4}{t+1} dt \\
 &= -[2t - 4\log(t+1)]_0^1 + [2t - 4\log(t+1)]_1^7 \\
 &= -(2 - 4\log 2 - 0 + 4\log 1) + (14 - 4\log 8 - 2 + 4\log 2) \\
 &= -2 + 4\log 2 + 14 - 4\log 8 - 2 + 4\log 2 \\
 &= 10 + 8\log 2 - 4\log 2^3 \\
 &= (10 + 8\log 2 - 12\log 2) \\
 &= (10 - 4\log 2)
 \end{aligned}$$

Question 16 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to find $\frac{dy}{dt}$, or equivalent merit	1

Sample answer:

$$y = 200(1 + 19e^{-0.5t})^{-1}$$

$$\text{Rate of growth} = \frac{dy}{dt}$$

$$= -200(1 + 19e^{-0.5t})^{-2} \times \left(\frac{-19e^{-0.5t}}{2} \right)$$

$$= \frac{1900e^{-0.5t}}{(1 + 19e^{-0.5t})^2}$$

Question 16 (b) (ii)

Criteria	Marks
• Provides correct solution with justification	2
• Provides range, or equivalent merit	1

Sample answer:

All terms in $\frac{dy}{dt}$ are positive so y is increasing.

$$\text{when } t = 0 \quad y = \frac{200}{1 + 19} = 10.$$

$$\text{so } y \geq 10 \text{ for } t \geq 0.$$

$$\text{As } t \rightarrow \infty \quad e^{-0.5t} \rightarrow 0$$

$$\text{so } y \rightarrow \frac{200}{1 + 0} = 200$$

$$\text{hence } 10 \leq y < 200.$$

Question 16 (b) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned}
 \frac{y}{400}(200 - y) &= \frac{1}{2(1 + 19e^{-0.5t})} \left(200 - \frac{200}{1 + 19e^{-0.5t}} \right) \\
 &= \frac{100}{(1 + 19e^{-0.5t})} \left(\frac{1 + 19e^{-0.5t} - 1}{1 + 19e^{-0.5t}} \right) \\
 &= \frac{1900e^{-0.5t}}{(1 + 19e^{-0.5t})^2} \\
 &= \frac{dy}{dt}
 \end{aligned}$$

Question 16 (b) (iv)

Criteria	Marks
• Provides correct solution	2
• Recognises the importance of the vertex of the parabola in part (iii), or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{dy}{dt} &= \frac{y}{400}(200 - y) \\
 &= \frac{200y - y^2}{400}
 \end{aligned}$$

which is a quadratic in y with roots at $y = 0$ and $y = 200$. Since the coefficient of y^2 is negative, the quadratic has a max at $y = 100$.

$\therefore$ Population growing fastest when population is $y = 100$.

2016 HSC Mathematics

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	4.2	P5
2	1	3.1	H5
3	1	9.1, 9.5	P5
4	1	5.1	P4
5	1	12.5, 13.5	H5
6	1	13.3	H5
7	1	13.1	H5
8	1	1.2, 13.2	H5
9	1	1.2, 11.1	H8
10	1	12.2	H3

Section II

Question	Marks	Content	Syllabus outcomes
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11 (b)	2	8.9	P7
11 (c)	2	1.2	P3
11 (d)	2	11.1	H5
11 (e)	3	1.4, 6.3, 9.1	P4
11 (f)	2	1.1, 8.4, 13.5	P6, H5
11 (g)	2	13.1, 13.2	H5
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12 (a) (ii)	2	6.5	P4
12 (a) (iii)	2	2.3, 6.5	P4
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13 (a) (i)	4	10.2, 10.4	H6
13 (a) (ii)	2	10.5	H6
13 (b) (i)	2	1.3, 9.5	P5
13 (b) (ii)	1	9.5	P5
13 (c) (i)	1	14.2	H4
13 (c) (ii)	2	14.2	H4, H5

Question	Marks	Content	Syllabus outcomes
13 (d)	3	11.4, 13.6	H8
14 (a)	3	11.3	H5, H8
14 (b) (i)	2	7.5	H4, H5
14 (b) (ii)	1	7.5	H4, H5
14 (b) (iii)	1	7.5	H4, H5
14 (c) (i)	1	10.6	H4, H5
14 (c) (ii)	3	10.6	H4, H5
14 (d)	2	7.2, 8.2	P8, H5
14 (e)	2	7.1, 12.2	H3, H5
15 (a)	4	11.4	H8
15 (b) (i)	2	3.3	H5
15 (b) (ii)	3	3.3, 7.2, 12.2	H3, H5
15 (c) (i)	2	2.3, 2.5	H5
15 (c) (ii)	2	2.3, 2.5	H5
15 (c) (iii)	2	2.3, 2.5	H5
16 (a) (i)	1	14.3	H4
16 (a) (ii)	2	14.3	H4, H5
16 (a) (iii)	2	4.2, 10.5	H4, H5, H6
16 (a) (iv)	3	12.5, 14.3	H4, H5
16 (b) (i)	2	12.5, 14.1	H5
16 (b) (ii)	2	4.1, 12.5	H5
16 (b) (iii)	1	1.3, 14.2	H3
16 (b) (iv)	2	9.1, 12.5, 14.2	H5, H7